

Department of Computer Science and Engineering

Digital Electronics (ESCS-101)

Assignment - 1

- Last date to submit: 19/09/2025

Sr. No.	Question	Course Outcome	RBT Level
1	a) Evaluate the value of x in following: $(85.63)_{10} = (X)_2$ $(300.45)_{10} = (X)_8$ $(F3A7C2)_{16} = (X)_2$ $(1101.101)_2 = (X)_{10}$ $(1101010)_2 = (X)_{\text{gray}}$ b) Interpret the decimal numbers (a) 27, (b) 396, and (c) 4096 in binary form, ASCII code, Gray Code and Excess 3 codes. c) Noting that $3^2=9$, formulate a simple procedure for converting base-3 numbers directly to base 9. Further use the procedure to convert $(2110201102220112)_3$ to base 9 d) Perform arithmetic operations $(+42) + (-13)$ and $(-42)-(-13)$ in binary using the assigned 2's complement representation of negative numbers.	CO1	L5, L6
2	a) A certain system contains two identical circuits in parallel. As long as both are operating properly, the outputs of both circuits are always the same. If one of the circuits fails, the outputs will be at opposite levels at some time. Devise a way to detect that a failure has occurred in one of the circuits. b) Reduce the following Boolean expression to the indicated number of literals: $A'C' + ABC + AC'$ to 3 literals. $A'B(D' + C'D) + B(A + A'CD)$ to 1 literal. $(A' + C)(A' + C')(A + B + C'D)$ to 4 literals.	CO2	L5
3	Convert the following into standard SOP and POS form.	CO2	L5

	(i) $(AB + C)(B + C'D)$ (ii) $X' + X(X + Y')(Y + Z')$ (iii) $B'D + A'D + BD$ (iv) $(XY + Z)(XZ + Y)$ Simplify using K maps. (i) $F(W, X, Y) = \Sigma m(1, 4, 5, 6)$ (ii) $AB'C + B'C'D + BCD + ACD' + A'B'C + A'BC'D$ (iii) $F(A, B, C, D, E) = \Sigma m(0, 2, 3, 4, 5, 6, 7, 11, 15, 16, 18, 19, 23, 27, 31)$ (iv) $F(A, B, C, D) = \Pi M(0, 1, 2, 3, 4, 10, 11)$		
4	(a) Draw the TTL circuit with Totem pole output and verify that it performs NAND gate operation. (b) A TTL gate has the following actual voltage level values: $V_{IH}(\min) = 2.25V$, $V_{IL}(\max) = 0.65V$. Assuming it is being driven by gate with $V_{OH}(\min) = 2.4V$, $V_{IL}(\max) = 0.40V$, analyze whether the gate's HIGH- and LOW-level noise margins are to be suitable for a circuit design or not.	CO3	L5 L4